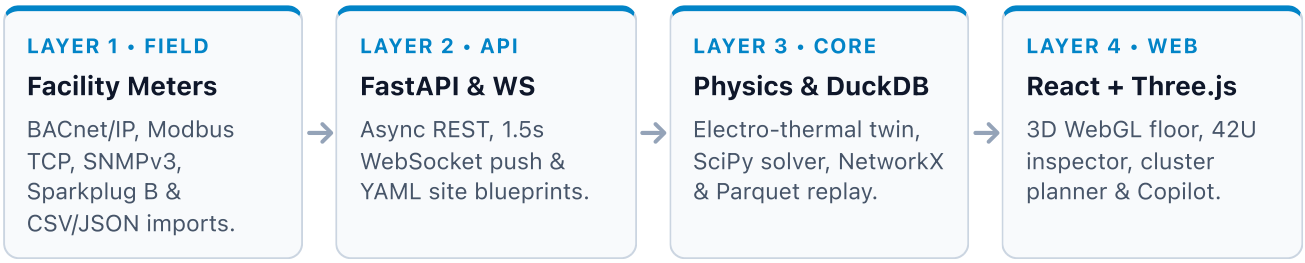
Pandita's Aether-DCTWIN

Technology Stack & System Architecture Reference

1. End-to-End Architecture



> **Strict Facility-Only Boundary:** Runs on-prem in the facility network with zero agents on customer servers, zero OS/BMC credentials, and zero workload throttling.

2. Frontend & 3D Visualization Stack

TECHNOLOGY	HOW IT IS USED IN AETHER-DCTWIN
React 18 & TypeScript	Strongly typed single-page workspace (`Overview` , `Inspect` , `Plan` , `Health` , `ROI` , `Copilot`) with live WebSocket state sync.
Three.js & React Three Fiber	Hardware-accelerated 3D WebGL data hall (`@react-three/fiber` , `@react-three/drei`) rendering rack heatmaps, overhead liquid coolant loops, airflow plumes, and click-to-inspect floor equipment.
Vite 5 & Dynamic Plugins	Fast production bundling plus conflict-free UI plugin discovery (`import.meta.glob`) so engineers add feature tabs independently.
Tailwind CSS & Recharts	Clean Dark/Light operational design system with tabular numeric alignment (`tabular-nums`) and interactive PUE / 30-minute forecast charts.

3. Backend, Streaming & Data Historian Stack

TECHNOLOGY	HOW IT IS USED IN AETHER-DCTWIN
Python 3.11+, FastAPI, Uvicorn	High-throughput async REST API, automatic OpenAPI/Swagger documentation (`/docs`), and real-time `/ws/telemetry` push server.
Pydantic v2 & PyYAML	Validates all API contracts and loads declarative facility blueprints (`configs/sites/*.yaml`) so any row/rack/CDU/tariff layout runs without code edits.
DuckDB & Apache Parquet	Embedded columnar historian (`duckdb` + `pyarrow`) storing 24-hour facility snapshots for sub-10ms DVR time-travel scrubbing and SQL analytics—zero external DB setup.
NetworkX	Models the directed electrical & cooling topology graph (`UPS → PDU → Rack ← CDU / CRAC`) for blast-radius checks and maintenance checklists.

4. Physics, Control & Applied AI Stack (Why Each Tool Was Chosen)

First-Principles Physics SDK `backend/domain/`

Pure, unit-tested equations for Magnus-Tetens dew point, chiller COP & free-cooling curves, cubic fan/pump affinity laws, sensible coolant flow, and 3-phase AC neutral current phasors.

Zone Self-Calibration NumPy (RLS Filter)

Online Recursive Least Squares filter continuously learns zone thermal resistance and capacitance from PDU & cooling meters to flag strainer and heat-exchanger fouling early.

Constrained Cooling Control SciPy (SLSQP MPC)

Solves 30-minute lookahead cooling trajectories using sequential quadratic programming under strict ASHRAE inlet ceilings (27°C) and hardware fan/pump ramp-rate limits.

Multi-MW AI Cluster Planner cluster_planner.py

Sizes 2k–25k GPU buildouts (`GB200 NVL72` , `HGX H200`) across multi-hall pods, 87/13 liquid-to-air heat splits, MV transformers, and <100ms training step-load BESS buffers.

Physics-Guided Anomaly Detection scikit-learn

Pairs `IsolationForest` outlier scoring with thermodynamic energy-balance residuals to distinguish genuine fan trips or damper leaks from normal AI workload surges.

Tool-Calling Facility Copilot OpenRouter / OpenAI

Function-calling LLM loop (with offline deterministic fallback) wired to 7 live inspection, DuckDB SQL, and simulation tools so every number is auditable—never hallucinated.

5. Field Integration, Team Modularity & Deployment

Field Protocol Safety

- **Pluggable Adapter Layer:** Clean interface separating simulation from live BACnet/IP, Modbus TCP, SNMPv3 & Sparkplug B gateways.
- **Automatic BMS Fallback:** Supervisory writes use BACnet Slot 8 and clear cleanly back to local Slot 16 defaults whenever Auto-Pilot is turned off.
- **BYOD CSV/JSON Mapper:** Instant offline evaluation on customer historian exports.

Parallel Team Velocity

- **Zero-Conflict Plugins:** Auto-mounted FastAPI routers (`api/features/*.py`) and React tabs let engineers build features without touching shared files.
- **Config-Driven Sites:** Customer topologies, equipment ratings & utility tariffs live in declarative YAML.
- **SWE Domain Onboarding:** Built-in Critical Facilities Rosetta Stone for non-MEP software engineers.

CI/CD & Deployment

- **Automated Guardrails:** `Pytest + GitHub Actions` CI verify physics laws, multi-site configs, and strict zero-server-access rules on every PR.
- **Single-Artifact Runtime:** FastAPI serves both the REST/WS backend and compiled React/WebGL bundle.
- **Container Ready:** Ships with `Dockerfile` , `docker-compose.yml` , and 1-command local launcher.